

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week -3

TASK-1

STAGE INPUT

OPTIONS ▾

Sample of 4 documents

```
_id: ObjectId('6aae08f8463ca...  
title : "The Silent  
        Algorithm"  
author : "R. Menon"  
genre : "Technology"  
publishedYear : 2022  
▸ formats : Array (3)  
▸ specs : Object (5)  
  avgRating : 4.6  
▸ reviews : Array (2)
```

```
_id: ObjectId('6aae0905463ca...  
title : "Ocean of Code"  
author : "A. Fernandes"  
genre : "Technology"  
publishedYear : 2021  
▸ formats : Array (2)  
▾ specs : Object (3)  
  pages : 210  
  fileSize_MB : 2.1  
  drm : false  
  avgRating : 4.2  
▸ reviews : Array (2)
```

```
_id : ObjectId('6aae0910463ca...')
title : "Whispers of
        Yesterday"
author : "S. Kapoor"
genre : "Fiction"
publishedYear : 2019
formats : Array (2)
specs : Object (2)
avgRating : 3.8
reviews : Array (2)
```

```
_id: ObjectId('6aae0923463ca...')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
formats: Array (2)
specs: Object (4)
  avgRating: 4.9
reviews: Array (2)
```

TASK-2

🕒

{ "avgRating": { "\$gt": 4 } }

Explain

Reset

Find

Options ▶

+

📄

🔗

</>

25

1 - 3 of 3

↺

↻

↷

⋮

{ }

🗑️

🏠 book_metadata

	_id ObjectId	title String	author String	genre String	
1	ObjectId('6aae0923463ca1e...')	"Quantum Horizons"	"R. Menon"	"Science"	✎ 📄 📄 🗑️
2	ObjectId('6aae08f8463ca1e...')	"The Silent Algorithm"	"R. Menon"	"Technology"	✎ 📄 📄 🗑️
3	ObjectId('6aae0905463ca1e...')	"Ocean of Code"	"A. Fernandes"	"Technology"	✎ 📄 📄 🗑️

🕒

{ "formats": { "\$in": ["Digital", "Audiobook"] } }

Explain

Reset

Find

Options ▶

+

📄

🔗

</>

25

1 - 3 of 3

↺

↻

↷

⋮

{ }

🗑️

🏠 book_metadata

	_id ObjectId	title String	author String	genre String	
1	ObjectId('6aae08f8463ca1e...')	"The Silent Algorithm"	"R. Menon"	"Technology"	✎ 📄 📄 🗑️
2	ObjectId('6aae0923463ca1e...')	"Quantum Horizons"	"R. Menon"	"Science"	✎ 📄 📄 🗑️
3	ObjectId('6aae0905463ca1e...')	"Ocean of Code"	"A. Fernandes"	"Technology"	✎ 📄 📄 🗑️

```
{
  "formats": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

[Explain](#) [Reset](#) [Find](#) [Options ▶](#)

+ 📄 🔗 </>
25 1 - 3 of 3 ↺ ↻ ⌵ ☰ | { } 🏠

book_metadata				
	_id ObjectId	title String	author String	genre String
1	ObjectId('6aae08f8463cale...)	"The Silent Algorithm"	"R. Menon"	"Technology"
2	ObjectId('6aae0923463cale...)	"Quantum Horizons"	"R. Menon"	"Science"
3	ObjectId('6aae0905463cale...)	"Ocean of Code"	"A. Fernandes"	"Technology"

```
{
  "reviews.comment": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

+ 📄 🔗 </> 💡
25 1 - 3 of 3 🔄 < > ⌵ ☰ | { } 🏠

book_metadata				
	_id ObjectId	title String	author String	genre String
1	ObjectId('6aae08f8463cale...	"The Silent Algorithm"	"R. Menon"	"Technology"
2	ObjectId('6aae0905463cale...	"Ocean of Code"	"A. Fernandes"	"Technology"
3	ObjectId('6aae0923463cale...	"Quantum Horizons"	"R. Menon"	"Science"

TASK-3

```
> db.book_metadata.aggregate([ { $match: { publishedYear: { $gt: 2020 } } }, { $unwind:
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
}
BookFlow > |
```

TASK-4

>_MONGOSH



```
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 })
< avgRating_-1
> db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats")
< {
```

```
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
  },
  indexFilterSet: false,
  queryHash: '85BA8276',
  planCacheShapeHash: '85BA8276',
  planCacheKey: '6EB4A3FD',
  optimizationTimeMillis: 27,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
```

```
stage: 'FETCH',
inputStage: {
  stage: 'IXSCAN',
  keyPattern: {
    avgRating: -1
  },
  indexName: 'avgRating_-1',
  isMultiKey: false,
  multiKeyPaths: {
    avgRating: []
  },
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: {
    avgRating: [
      '[inf.0, 4)'
    ]
  }
},
rejectedPlans: []
},
executionStats: {
```

```
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 28,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3
```

```
advanced: 3,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
docsExamined: 3,
alreadyHasObj: 0,
inputStage: {
  stage: 'IXSCAN',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  keyPattern: {
    avgRating: -1
  },
  indexName: 'avgRating_-1',
  isMultiKey: false,
  multiKeyPaths: {
```

```
      avgRating: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      avgRating: [
        '[inf.0, 4)'
      ]
    },
    keysExamined: 3,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0
  }
}
},
queryShapeHash: '79E1597FBF1E7E0551896CC4A5AA0604E417E5E37F937A039ABC2D303FB513A',
command: {
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  }
}
```

>_MONGOSH



```
    '$gt': 4
  }
},
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'Hemaswi',
  port: 27017,
  version: '8.0.32',
  gitVersion: 'f9eb55a7cc900f33a722a5b32a0fdbf54385c749'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
```

BookFlow>